AI Contribution Log

Course: 02AML204 – Introduction to Artificial Intelligence

PRN: 25UAM061 | **Name:** Atharv Deepak Shinde | **Division:** A

Contribution 1 (SLE-1)

AI Tool: GitHub Copilot

Task: Add functionality for entering study hours in the Study Time Tracker (C).

Prompt: Create beginner-friendly C code that asks the user how many hours they studied and stores the value in a variable.

AI Contribution: Copilot suggested the variable and user-input code.

My Action: I reviewed the suggestion, understood the use of `scanf()`, and tested the code.

Testing: I entered 3 hours and checked the output.

Result: Successful.

Contribution 2 (SLE-2)

AI Tool: Claude (Anthropic)

Task: Implement BFS and A* for the 8-puzzle and the profiling script.

AI Contribution: Wrote the BFS / A* Python code (including the Manhattan-distance heuristic) and the timing and node-counting script, and drafted the report structure.

My Action: Reviewed the puzzle setup and code logic, ran the profiling script, and checked that node counts and timings made sense.

Testing: Ran both algorithms on the same start and goal state, 3 runs each. Both returned the same 8-move solution. BFS expanded 201 nodes (0.3124 ms) and A* expanded 11 nodes (0.0613 ms).

Result: Successful. I wrote the justification from my own numbers.

Contribution 3 (SLE-3)

AI Tool: Claude (Anthropic)

Task: Create the C4 architecture model (Context, Container, Component, Code).

AI Contribution: Explained the C4 requirements, suggested the container/component breakdown, drafted the diagrams and structured the report.

My Action: Chose the system, reviewed the diagrams against my own code, and checked that the design matched what I implemented.

Result: Successful.

Contribution 4 (SLE-4)

AI Tool: Claude (Anthropic)

Task: Write the Architecture Decision Record (ADR) and viva preparation notes.

AI Contribution: Structured the ADR in the required format, tidied the wording, and suggested possible viva questions.

My Action: Chose the decision (A* over BFS), supplied the SLE-2 profiling numbers, checked that the figures match my report, and reviewed the trade-offs so I can explain them in the viva.

Result: Successful.

Summary

AI helped with code drafts, structure and wording. I made the design decisions, ran the experiments, verified the results and can explain every part of my work.